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Financial Analytics Lab

GROUP 11

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02

Answers

[Link to PYTHON CODE Q1]
[Link to PYTHON CODE Q2,3]

Q1. Approach in Python:

Outcome: Deriving returns based on closing prices, we analyzed the data.

Day-of-the-week Extraction: We isolated and utilized the day component from the dataset for grouping purposes.

Statistical Examination: The returns were grouped by day, and statistical measures, including F-statistic and P values, were computed.

GARCH Modeling: Adhering to the methodology outlined in the paper, we implemented a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model to explore volatility variations across distinct weekdays.

1. BSE

For Investors: While Tuesdays show a promising average return of 0.32% compared to the less favorable -0.12% on Mondays, the overall test with a p-value of 0.154 suggests that this pattern lacks sufficient strength for reliable investment decision-making. Relying solely on the day of the week may not be a prudent strategy.

Market Dynamics: These subtle trends, such as the better performance on Tuesdays, challenge the notion of complete unpredictability in stock prices. However, given the lack of robust evidence indicated by the p-value, it doesn't serve as a warning for the market; rather, it remains an intriguing observation.

Risk Considerations: The study reveals that stock price volatility doesn't exhibit a preference for any particular day, remaining relatively consistent throughout the week. This implies that there is no inherently "safest" day for buying or selling if the goal is to mitigate risks.

2. ONGC

Returns: Tuesday appears slightly more favorable for returns, standing at around 0.50%, while Monday also holds a respectable figure of 0.44%. However, the test results, with a p-value of 0.628, indicate that these patterns lack reliability, rendering the day of the week inconsequential in predicting stock performance.

Risk: The volatility or "bumpiness" in the stock price doesn't show preferences throughout the week, even though Tuesday and Wednesday exhibit slightly elevated levels.

Answers

Q1. 3. NTPC

Returns: Tuesday stands out with an impressive 0.58% average return, surpassing all other days. Monday also shows positive results at 0.24%, but Thursday and Friday report losses. Despite these variations, the ANOVA test, indicating a p-value of 0.111, suggests that we cannot confidently assert that these differences are not merely random fluctuations.

Risk: The day-to-day fluctuation or "bumpiness" in NTPC's stock price remains relatively consistent throughout the week, with no significant variations in risk levels.

Implications: While Tuesday may seem enticing due to its higher returns, statistically, the observed difference is not substantial enough to rule out the possibility of chance. For NTPC, the day of trading likely won't have a significant impact in the long run. It's advisable not to place excessive reliance on the specific day when making trading decisions.

4. IOCL

Returns Overview: Tuesday shines with a nice bump in returns (0.47%), and Wednesday also looks good (0.28%). On the flip side, Monday's a bit down (-0.19%). The ANOVA test, with a p-value of 0.302, tells us these differences in returns throughout the week aren't statistically solid.

Volatility Check: Looking at how much IOCL's stock prices swing, there's not much difference from one day to the next. Tuesday and Wednesday are slightly higher in volatility, but it's pretty close across all days.

Simple Takeaway: IOCL has its ups and downs during the week, but there's no clear pattern that says one day is best for trading. Even though Tuesday and Wednesday have higher returns, the stats aren't strong enough to make a trading plan around it. Keep your strategy flexible and not just tied to the calendar.

5. GAIL

Returns Analysis: Tuesday demonstrates a distinctive performance with an average return of 0.27%, and Monday also exhibits positive returns at 0.18%. Conversely, Friday experiences a decline with a negative return of -0.26%. However, the ANOVA test produces a non-significant p-value of 0.621, suggesting that these daily return differentials lack statistical significance.

Risk Assessment: The fluctuation in GAIL's stock prices demonstrates relatively uniform patterns across the days of the week. Although Wednesday displays slightly higher volatility, overall, there is minimal variance in volatility throughout the week.

Practical Insight: While GAIL exhibits some fluctuations in daily returns, notably with Tuesday as a potential highlight, the statistical assessment indicates that this observed pattern lacks reliability for actionable trading strategies. Similarly, volatility does not significantly vary with the day of the week. Consequently, for GAIL, the chosen trading day does not exhibit a substantiated impact on returns; broader market conditions and other factors play a more decisive role than the specific day of the week.

Answers

Q2. Model Evaluation:
The comprehensive analysis of the multiple regression output sheds light on the model's robustness and its ability to explain variations in the dependent variable, y.

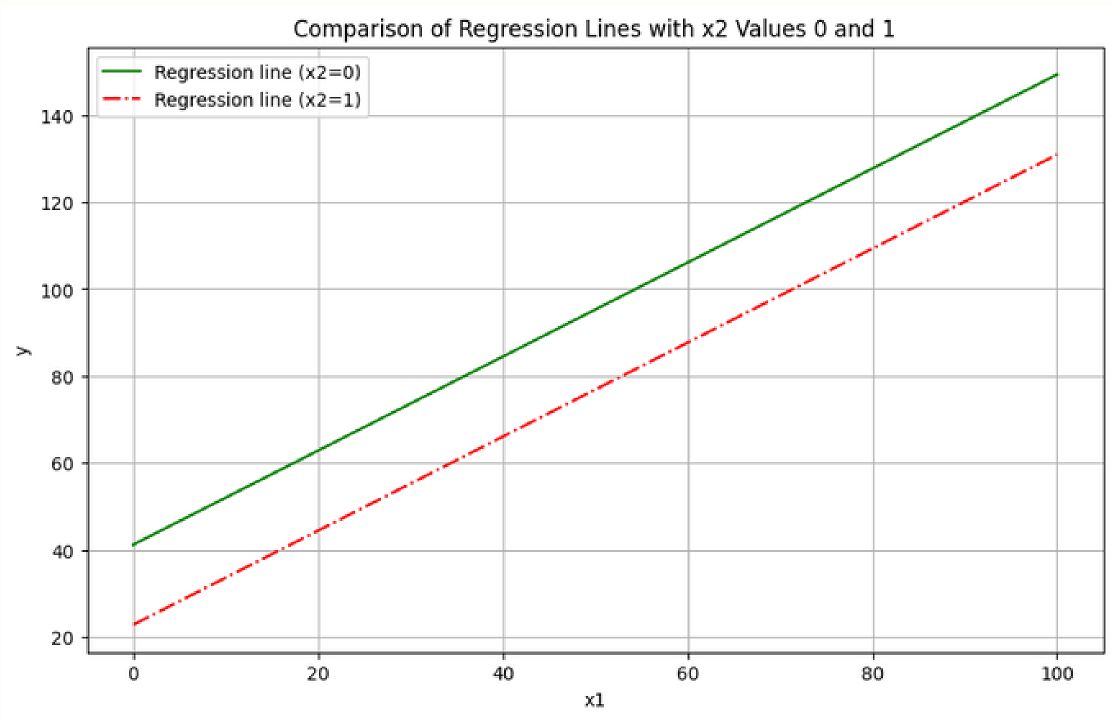
1. Model Strength:

- Multiple R: The Multiple R, standing at 0.623, portrays a moderate positive correlation between the independent variables (x1 and x2) and the dependent variable (y).
- R Square: An R Square value of 0.388 implies that approximately 38.8% of the variability in y can be elucidated by the predictors. The Adjusted R Square of 0.341, considering the model's complexity, still signifies a commendable 34.1% explanatory power.
- Standard Error: The Standard Error of 11.744 serves as a measure of the average deviation of observed values from the regression line, amounting to 11.744 units.
- F-statistic: A statistically significant F-statistic (p-value = 0.0017) underscores the overall significance of the model, indicating that at least one of the coefficients significantly influences y.

2. Dummy Variable Contribution:

- The dummy variable x2 exhibits a substantial impact on the model. With a coefficient of -18.404, a shift from x2 = 0 to x2 = 1 is associated with an anticipated decrease in y by 18.404 units, holding other variables constant.
- The low p-value of 0.0004 reinforces the significance of x2 in predicting y, surpassing the conventional alpha level of 0.05.

Analyzing Dummy Variable Influence:
To further grasp the nuances of the dummy variable's contribution, plotting the regression line for y against x1 under different conditions of x2 (0 and 1) is paramount. This visual representation will elucidate the distinct impact of the dummy variable, showcasing how the regression line adjusts vertically with the change in x2, providing a tangible insight into the variable's effect on the model.



Answers

Q3. OLS Regression Results

Dep. Variable:	y	R-squared:	0.873
Model:	OLS	Adj. R-squared:	0.834
Method:	Least Squares	F-statistic:	22.33
Date:	Thu, 07 Mar 2024	Prob (F-statistic):	1.00e-05
Time:	10:44:50	Log-Likelihood:	-28.725
No. Observations:	18	AIC:	67.45
Df Residuals:	13	BIC:	71.90
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
intercept	13.0257	1.734	7.513	0.000	9.280	16.771
x1	-2.4918	0.839	-2.970	0.011	-4.305	-0.679
x2	3.3956	0.907	3.745	0.002	1.437	5.355
x3	-5.0401	0.865	-5.825	0.000	-6.909	-3.171
x4	-2.8760	0.998	-2.882	0.013	-5.032	-0.720

Omnibus:	5.013	Durbin-Watson:	2.131
Prob(Omnibus):	0.082	Jarque-Bera (JB):	2.599
Skew:	-0.626	Prob(JB):	0.273
Kurtosis:	4.377	Cond. No.	13.2

Answers

Q3. OLS Regression Analysis Results

The presented output unveils insights derived from an Ordinary Least Squares (OLS) regression model featuring the following attributes:

- Dependent Variable: y
- Independent Variables: x1, x2, x3, x4
- R-squared: 0.873
- Adjusted R-squared: 0.834
- Number of Observations: 18

Interpretation:

- The formidable R-squared (87.3%) denotes that a substantial portion of the variance in the dependent variable y finds explanation within the model.
- The Adjusted R-squared (83.4%), factoring in the number of predictors, provides a nuanced metric for model comparison.

Independent Variables:

- x1: A unit increase in x1 forecasts a decrease in y by 2.4918, holding other variables constant (statistically significant, $p < 0.05$).
- x2, x3, x4: These depict categories of the second independent variable. The positive coefficient for x2 suggests a positive effect on y, while x3 and x4 indicate negative effects.

Dummy Variables:

- The indicator variable encompasses 4 categories in total, with x1 serving as the reference.
- p-values for x2, x3, and x4 assert their statistical significance as predictors of y, except for x4 which is marginally significant ($p = 0.013$).

Model Fit:

- A substantial F-statistic with a low probability (1.00e-05) underscores the model's excellence in capturing the data.
- The Durbin-Watson statistic (2.131) suggests the absence of serious autocorrelation in residuals.

Assumptions Checking:

- Omnibus (5.013) and Prob(Omnibus) (0.082): Residuals might deviate from normality, though the p-value exceeds 0.05.
- Skewness (-0.626): Residual distribution may exhibit a slight leftward skew.
- Kurtosis (4.377): Residual tails could be heavier than a normal distribution (leptokurtic).
- Jarque-Bera (JB) and Prob(JB): Normality tests yield a JB value of 2.599 and a p-value of 0.273, failing to reject the null hypothesis at the 5% level.
- The Condition Number (13.2): Signals a moderate level of multicollinearity, generally not alarming.

In conclusion, the model showcases strong predictive capabilities with significant variables. However, the assumption of normally distributed residuals warrants further exploration through residual plots and additional diagnostics beyond the provided output.

The total mumber of categories as portrayed from the data provided are 4 namely,
(1,0,0)
(0,1,0)
(0,0,1)
(0,0,0)
Total 3 variables are enough to explain this categorical classification for the second independent variable.